Enhancing Brain Tumor Classification Through Image Subtraction Analysis

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***Abstract*—Local Binary Pattern (LBP) is widely used for categorizing brain tumor. Nonetheless, LBP is impacted strongly by the spurious noise. Besides, it employs a constant neighborhood size that does not accurately represent textures of different sizes or orientations. To improve the classification accuracy and overcome these shortcomings, we have utilized LBP to extract texture characteristics from brain MRI images and have generated a texture contrast map by subtracting the original image from the LBP image. This map effectively amplifies the unique textures linked to tumor regions, reducing some of the limitations of standard LBP and ultimately enhancing the precision of the brain tumor classification system. The effectiveness of our approach is demonstrated by a classification accuracy of 95.44%. This result underscores the potential of the enhanced LBP-based method, combined with k-nearest neighbours (k-NN) classification, for brain tumor categorization.**

***Index Terms*—brain tumor classification, local binary pattern, texture feature extraction, k-nearest neighbors, data augmentation**

I. Introduction

The field of medical diagnostics has seen great advancements in recent years, driven by both emerging technologies and improved methods for accurately categorizing and identifying diseases. Among these challenges, tumor remain one of the most difficult medical problems, characterized by the excessive growth of abnormal cells within the brain. Such growth not only damages healthy brain tissues but also affects nearby critical structures such as nerves and the pituitary gland. tumor exert constant pressure on vital brain regions, leading to serious and often life-threatening effects [1]. tumor are generally classified into two main categories: primary and secondary. A primary tumor originates within the brain itself and can be either malignant or benign. In contrast, a secondary tumor develops when cancer cells spread to the brain from another part of the body. Primary tumor include several types, such as pituitary adenoma, glioma, and meningioma. Proper classification and detection of tumor is essential for effective treatment planning. Researchers have explored various methods to enhance the diagnosis and classification of these tumors.

Medical image analysis has undergone a major transformation in algorithm design and formulation, driven by the integration of machine learning (ML) and advanced image processing techniques. The k-NN method is a simple yet powerful ML algorithm that has been widely used [2]. The method performs well in classification and regression tasks, especially with datasets containing intricate decision boundaries. The adoption of such ML systems has significantly improved diagnostic accuracy, making them vital tools for clinical decision-making. Another key technology in this area is the LBP, which is primarily used as a feature extraction method to represent medical images. The major drawback of LBP is that it can introduce random noise, which adversely impacts classification accuracy. Most brain MR images exhibit uniform patterns, and LBP tends to perform poorly in such cases. Several variants of LBP have been proposed in the literature to improve its classification capability, often in combination with k-NN for tumor classification [3].

In this paper, we focus on improving the classification accuracy of tumor detection, where the LBP of the MR image is subtracted from the original image, and the resulting image is used as input to the k-NN classifier. The model is trained and tested to verify its reliability. The proposed model is evaluated using various performance metrics, supported by a confusion matrix. The simulation results show that the proposed approach provides better precision and test accuracy than existing methods.

The remaining sections of the paper is organized as follows. Section II discusses the related research on tumor classification based on various promising approaches. The methodology used in this work is described in Section III. Section IV presents the performance evaluation and comparison. Finally, Section V provides the concluding remarks on the work.

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II. Related Work

Various machine learning (ML) techniques such as Support Vector Machines (SVM), k-NN, logistic regression, and ensemble methods, as well as deep learning (DL) approaches like Convolutional Neural Networks (CNN), Transfer Learning (TL), and Recurrent Neural Networks (RNN), have been employed for tumor classification. Sachdeva et al. [4] have utilized SVM and Artificial Neural Networks (ANN) after performing segmentation, feature extraction, and feature reduction to classify tumors. Cheng et al. [5] have been able to improve the classification accuracy by incorporating data augmentation and partitioning techniques in tumor regions. Ismael et al. [6] have extracted statistical features and applied a backpropagation neural network for tumor classification using traditional ML algorithms. Gumaei et al. [7] have proposed a hybrid approach that combines multiple feature extraction methods with a regularized extreme learning machine to improve classification performance. Yin et al. [8] have developed a new classification approach using the whale optimization algorithm for improved tumor classification. Tahosin et al. [9] have emphasized systematically selecting the most discriminant features from MRI data to reduce dimensionality and remove redundant or noisy information. They have discussed the feature subset helps classifiers to learn more effectively and improve the tumor classification rate.

Raja et al. [10] have employed a hybrid approach combining deep autoencoders with Bayesian fuzzy clustering for tumor classification. Abiwinanda et al. [11] have utilized CNN to classify tumors, while Pashaei et al. [12] have integrated CNNs with extreme learning machines to enhance classification accuracy. Ozyurt et al. [13] have developed a tumor detection system by applying a CNN with neutrosophic expert maximum fuzzy-sure entropy, showcasing the integration of diverse computational intelligence methods. In a subsequent study, Ozyurt et al. [14] have introduced a detection framework combining fuzzy C-means clustering, image super-resolution, CNNs, and extreme learning machines, effectively merging conventional and deep learning approaches for improved precision. Kaplan et al. [15] have applied two distinct feature extraction techniques, namely the distance-based LBP (distance-LBP) and angular-LBP to classify glioma, meningioma, and pituitary. They have employed classification algorithms including k-NN, ANN, Random Forest (RF), and Linear Discriminant Analysis (LDA) on distance-LBP, angular-LBP, and traditional LBP features. Their results have shown that the n-LBP method (with distance = 1) combined with a k-NN classifier has achieved the best classification accuracy of 95.56%.

III. Proposed Classification Model

In this Section, we discuss the datasets that we have used for the classification of tumor, along with the data partition. This is followed by a brief discussion on the exploration of the classification approach, and finally, the step-by-step procedure of the proposed methodology.

1) Datasets and Data Preparation: Two datasets of brain MR images are used in this study. One is available on the Figshare website [16], while the other dataset is available on the Kaggle website [17]. For convenience, we have named the Figshare dataset as DS-I and the Kaggle dataset as DS-II. The size of the images in both datasets is 512×512. Each dataset consists of brain MR images from three different classes: pituitary, meningioma, and glioma. The types of images in different classes of the datasets are specified in detail in [3]. Each dataset is organized into two subdirectories, where one contains files for training and the other contains files for testing and validation. Each subdirectory is further organized into three subdirectories, with each containing the image files of one class. The training subdirectory contains nearly 80% of the files, and the testing and validation subdirectory contains the remaining files. Table I shows the total number of brain MR images, as well as the number of images in different classes of the DS-I dataset. Similarly, Table II shows the number of images in different classes of the DS-II dataset.

A. Exploring a Model for the tumor Classification

In this work, we are trying to develop a classification model using LBP as a grayscale feature descriptor and k-NN as the ML tool for the classification of tumors based on the LBP features. LBP is not only sensitive to image noise, but also generates random noise, which can distort its texture descriptions. It inherently lacks scale and rotation invariance, reducing its reliability for analyzing objects of different sizes and orientations. To avoid noise, we consider using modified features by subtracting the image from LBP without using LBP directly as input to the classifier. The subtraction of the original image from the LBP-transformed image offers several benefits. It emphasizes significant texture details by reducing noise, leading to clearer and stronger texture patterns. Edge enhancement is also facilitated because LBP responds strongly to pixel value changes, and subtraction highlights boundaries, improving edge detection and subsequent feature extraction. Furthermore, this method helps to spotlight important features

TABLE I
Data Partition in DS-I Dataset

Sl. No.	Tumor Type	Total No. of Images	Training Images	Validation Images
1	Glioma	926	748	178
2	Meningioma	937	738	199
3	Pituitary	901	725	176

TABLE II
Data Partition in DS-II Dataset

Sl. No.	Tumor Type	Total No. of Images	Training Images	Validation Images
1	Glioma	1621	1271	350
2	Meningioma	1645	1330	315
3	Pituitary	1757	1412	345

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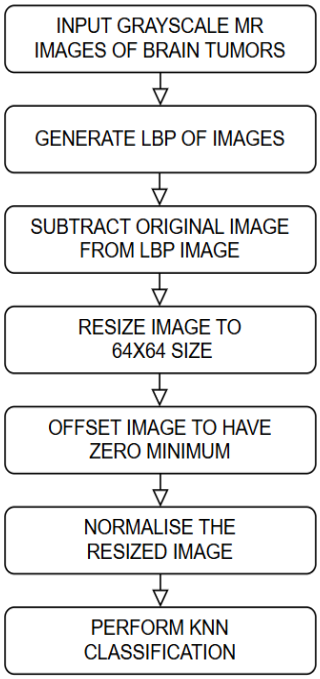

Fig. 1. Proposed Classification Framework

while suppressing irrelevant elements, enabling the system to focus on specific textural characteristics crucial for accurate analysis and classification. The subtracted images are resized to a lower resolution to reduce computational complexity. Finally, the scaled modified features are offset to make positive only values and scaled to have a consistent range, ensuring that all attributes contribute equally during the k-NN classification process.

B. Proposed Methodology

The proposed methodology consists of the following steps, and the step-by-step procedure of the proposed method of tumor classification is shown in Fig. 1.

1) **Grayscale Conversion:** All input images are first converted to grayscale. This transformation reduces the data complexity by eliminating color information, condensing each pixel to a single intensity value. This simplification decreases computational load and memory requirements, which is particularly beneficial as color channels are not essential for the texture-based analysis performed by the LBP operator.
2) **Primitive Feature Extraction:** Each image is partitioned into 3×3 non-overlapping blocks. The LBP operator is then applied to each block to convert the brain MRI images into LBP-transformed images, which highlight texture features.
3) **Image Subtraction:** The original grayscale images are subtracted from their corresponding LBP-transformed images. This operation aims to reduce noise and enhance contrast in the resulting subtracted images.
4) **Resizing:** All subtracted images are resized to a fixed dimension of 64×64 pixels to ensure dimensional consistency for model training. This standardization reduces computational load and helps mitigate overfitting.
5) **Offsetting:** After subtracting the original image from the LBP image, the resulting pixel values may be negative. To ensure the minimum value becomes 0, we apply an offset value $O = |\min(I)|$ to all pixels, where $\min(I)$ is the minimum value in the resulting image. If $\min(I)$ is negative, then $|\min(I)|$ is added to all pixel values. If $\min(I)$ is positive, then $\min(I)$ is subtracted from all pixel values. This adjustment guarantees that the new minimum value in the image is exactly 0.
6) **Normalization:** After offsetting, the pixel values are scaled to a standard range by dividing each pixel by $\max(I)$, the maximum pixel value in the image. This process, known as min-max normalization, maps all pixel values to the floating-point range [0, 1]. While the resulting image is not visually interpretable, it provides an ideal input for the classifier. This sequence enhances the stability and speed of model training by ensuring consistent intensity distributions and reducing variations caused by differences in lighting or contrast.
7) **Classification:** The normalized output of the resized images after the normalization is provided to the k-NN classifier. In this study, the choice of the parameter k was critical. We experimented with different values, such as $k = 3$, where the class is determined by a majority vote among the three closest neighbors. A lower k makes the model sensitive to local patterns and noise, while a higher k creates a smoother, but potentially less accurate, decision boundary when classes overlap.

IV. Results and Discussion

The visual outcomes of our proposed image subtraction model and the histogram for three different classes of tumors are given in Fig. 2. It can be observed that the subtraction of the image from the LBP image results in noise suppression. Moreover, the edges get more prominent in the subtracted image. The confusion matrix obtained from our experiment highlights the efficiency of the proposed classification system in identifying the type of tumor.

The confusion matrix for the classification in DS-I is given in Fig. 3. There are 159 correct predictions of glioma, 19 gliomas incorrectly classified, and no cases of glioma being falsely categorized as another class. In the meningioma category, there are 180 correct detections, 16 misclassifications, and 3 incorrect detections. The pituitary category has 175 correctly identified cases, 1 missed case, and none of them are incorrectly identified. We also have obtained the confusion matrix for DS-II dataset, which is shown in Fig. 4. We have obtained 332 true positives of correctly predicted glioma, 18 false negatives of glioma incorrectly classified as another class, and no false positives of glioma wrongly categorized as a different class. In the meningioma category, there are 297 correct detections, 14 missed detections, and 1 incorrect

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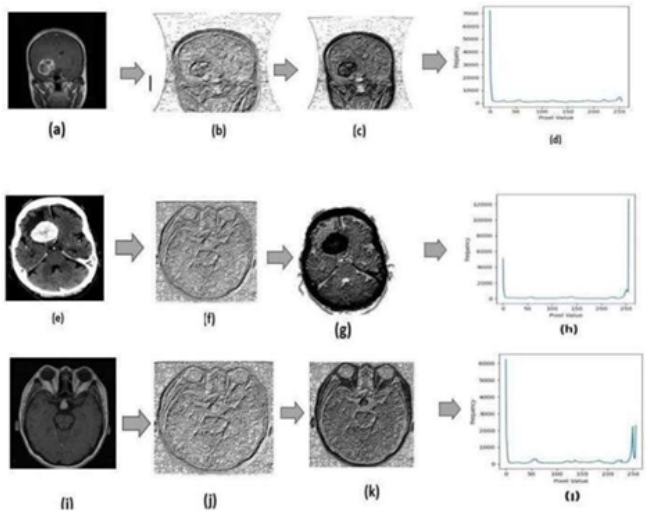

Fig. 2. (a) Original image of Glioma (b) LBP transformed image of Glioma (c) Subtracted Image of Glioma (d) Histogram of Glioma (e) Original image of Meningioma (f) LBP transformed image of Meningioma (g) Subtracted Image of Meningioma (h)Histogram of Meningioma (i) Original image of Pituitary (j) LBP transformed image of Pituitary (k) Subtracted Image of of Pituitary (l) Histogram of Pituitary.

detection. In the pituitary class, there are 342 correct positive predictions, 1 missed positive prediction, and no incorrect positive predictions.

The classification reports for our proposed method on both datasets are presented in Table III, Table IV respectively. For DS-I, the precision for glioma is 0.95954, meaning the model correctly identifies glioma cases nearly 96% of the time. The recall of 0.94857 indicates that approximately 94.86% of actual glioma cases are accurately detected. The F1-score, balancing precision and recall, stands at 0.95402. For meningioma, precision, recall, and F1-score are 0.93987, 0.95192, and 0.94586, respectively, demonstrating robust classification by the proposed model. The pituitary tumor class achieved impressive scores near 0.99708 across precision, recall, and F1-score, showing the excellent identification capability of the proposed model. Overall, our approach achieved a peak classification accuracy of 95.3% for tumor, highlighting its effectiveness in tumor classification. Table III represents the classification performance of the proposed model on the DS-II dataset. The model achieved a precision rate of approximately 90.86% in correctly identifying glioma cases, while the recall rate of about 89.33% indicates strong detection of actual glioma instances. The F1-score, which balances precision and recall. The proposed model achieves a 1.4% increase in real-time test accuracy compared to the standard LBP method in tumor classification. There are 100 instances of three types of tumor images used in the test operation.

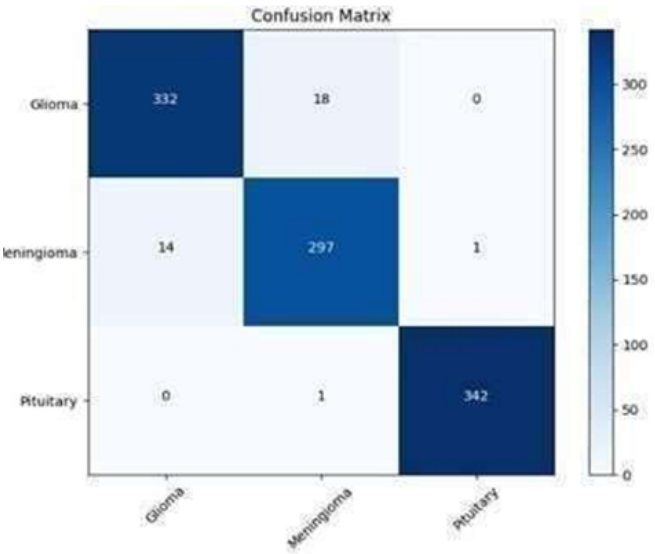

Fig. 3. Confusion Matrix on DS-I.

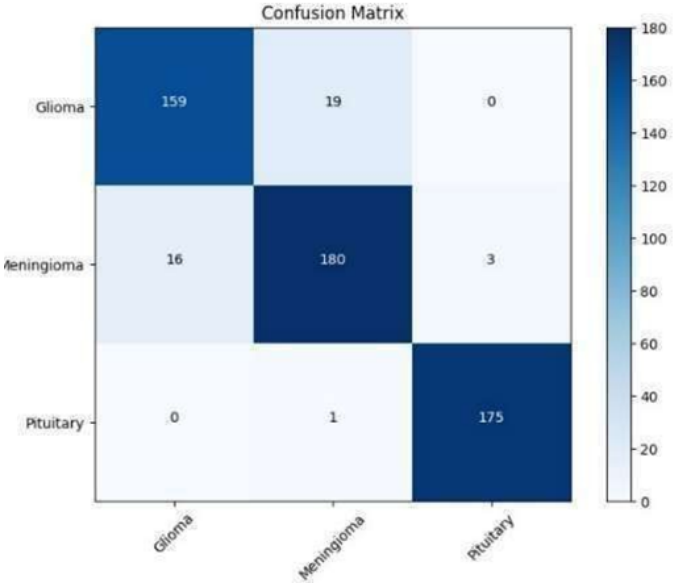

Fig. 4. Confusion Matrix on DS-II.

TABLE III
CLASSIFICATION PERFORMANCE ON DS-I. DATASET

Sl No.	Tumor Type	Precision	Recall	F1-score
1	Glioma	0.95954	0.94857	0.95402
2	Meningioma	0.93987	0.95192	0.94586
3	Pituitary	0.99708	0.99708	0.99708

TABLE IV
CLASSIFICATION PERFORMANCE ON DS-II DATASET

Sl No.	Tumor Type	Precision	Recall	F1-score
1	Glioma	0.95954	0.94857	0.95402
2	Meningioma	0.93987	0.95192	0.94586
3	Pituitary	0.99708	0.99708	0.99708

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V. Conclusions

A new method is proposed for tumor classification where the images are subtracted from their LBP representations. These resulting difference images provide unique features for discriminating between Glioma, Meningioma, and Pituitary tumors. We have analyzed the tumor feature histograms to understand the distribution of these features across the different tumor types. The subtracted images demonstrate lower noise, sharper edges, and improved contrast. For classification, a simple k-NN classifier was employed. The proposed approach outperforms conventional LBP-based methods, achieving an accuracy of 95.44%, which demonstrates its effectiveness. The classification accuracy could be further improved by exploring data augmentation and adaptive thresholding techniques.

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